ChronoGate

Standard Operating Procedure — interactive time-gating, lifetime & phasor FLIM analysis

Version 0.18.1

2026-08-05

PicoQuant .ptu · Becker & Hickl .sdt

1. Purpose & scope

This procedure describes how to operate **ChronoGate**, an interactive time-gating, rapid-lifetime and phasor viewer for PicoQuant `.ptu` (and Becker & Hickl `.sdt`) FLIM data. It covers loading data, setting the time gate, producing lifetime and phasor maps, running a rigorous IRF-reconvolution fit, and exporting results.

It is a practical operating guide, not a validation report. Absolute-lifetime work should use a measured IRF (see §10) and be cross-checked against a reference of known lifetime.

2. What ChronoGate does

A pulsed laser excites the sample; for every detected photon the hardware records the delay since the last pulse (the **microtime**). The per-pixel histogram of those delays is the **fluorescence decay**. ChronoGate reconstructs that decay cube and lets you:

- **Time-gate** — keep photons in a chosen microtime window and sum them into an image (fit-free lifetime *contrast*).
- **RLD** — read an apparent lifetime from two gates (fit-free, per pixel).
- **Phasor** — plot each pixel's decay as a point on the universal semicircle.
- **IRF fit** — recover a rigorous, IRF-deconvolved lifetime by reconvolution.

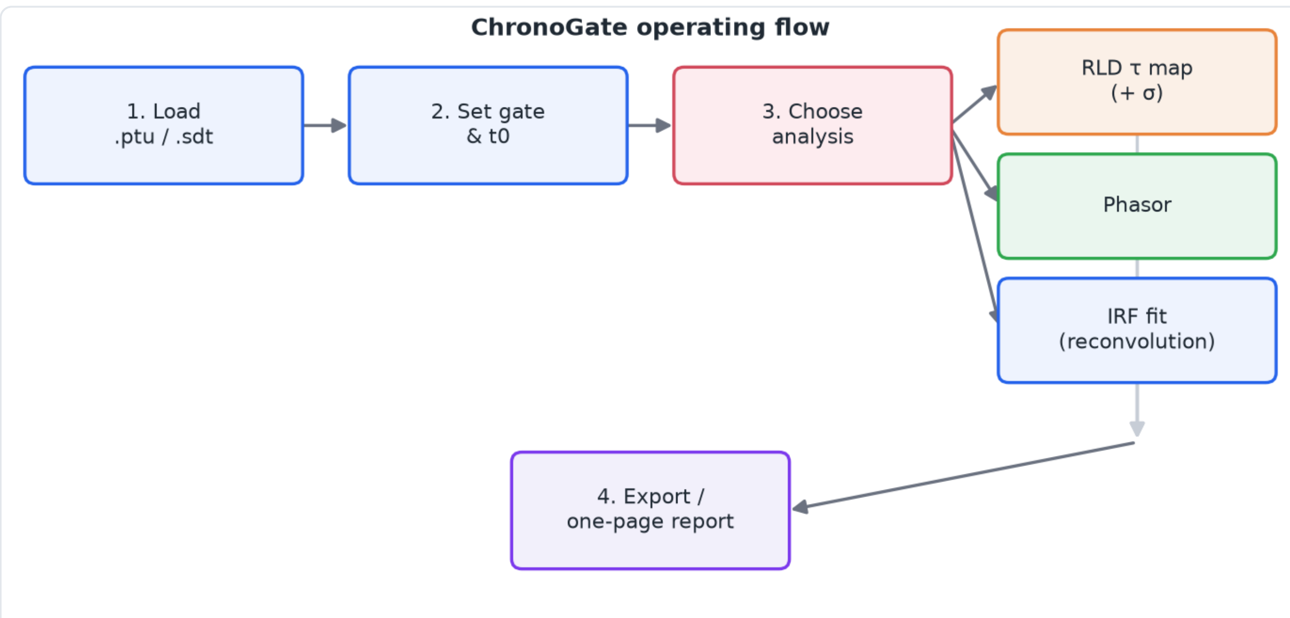

Figure 1. The end-to-end operating flow: load → gate → choose an analysis → export.

3. Install & launch

ChronoGate runs from source in a Python **3.12+** environment.

macOS (double-click launcher). Keep `ChronoGate.app` and `ChronoGate.command` together in the project folder and double-click `ChronoGate.app`. The first launch builds a virtual environment from `pyproject.toml` (a one-time download); later launches are instant. Changing dependencies triggers an automatic rebuild.

From a terminal (any OS).

```
python -m chronogate                 # opens a file picker
python -m chronogate path/to/file.ptu
python -m chronogate path/to/file.sdt
```

> If opening a `.sdt` reports that `sdtfile` is missing, or an IRF fit reports `scipy` is missing, your environment predates those dependencies — relaunch so the launcher rebuilds it, or run `pip install -e .`.
>
> **macOS keys:** shortcuts written `Ctrl+…` in this guide appear as **⌘ (Command)** on a Mac — e.g. `⌘O` to open, `⌘E` to export, `⌘R` for the report. The single-letter mode keys (`I` , `T` , `P`) are the same on every platform.

4. Opening data

- **File ‣ Open .ptu…** (`Ctrl+O`) — choose a `.ptu` or `.sdt` file.
- **File ‣ Open folder (stack)** — load a numbered z-series (e.g. `FLIM_stack_z1.ptu … z65.ptu`); step through planes with the **z-slice** slider in the File panel.
- **Channel** — for multi-detector files, pick the detector channel.

5. The workspace

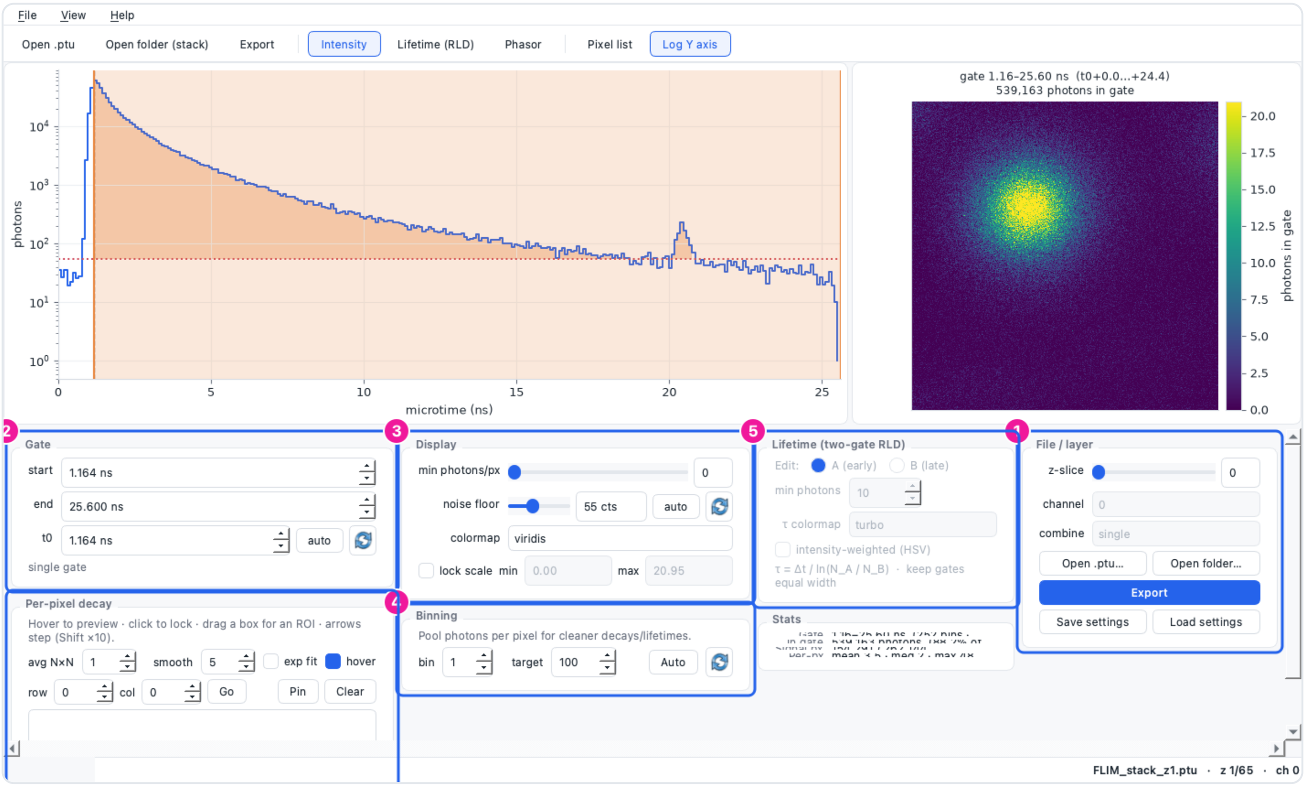

Figure 2. The main window. **1** File / layer (open, z-slice, channel, export, settings). **2** Gate (start, end, t0). **3** Display (threshold, noise floor, colormap, scale lock). **4** Binning. **5** Lifetime (RLD) controls. **6** Per-pixel decay / selection tools. The left plot is the summed decay with the gate shaded; the right plot is the current image.

Switch the image with the mode buttons in the toolbar: **Intensity** (`I`), **Lifetime (RLD)** (`T`), **Phasor** (`P`).

6. Setting the gate and t0

Figure 3. Gate panel: **1** gate start, **2** gate end (both in ns; you can also drag the shaded span on the decay plot), **3** t0 (pulse reference), **4** *auto* (set t0 to the smoothed decay peak), **5** *reset* (set t0 to its default, 0 ns).

1. Drag the shaded band on the decay plot, or type **start**/**end** in ns.
2. Set **t0** to the pulse arrival: click **auto** for the detected peak, or type a value. The **reset** (↻) button returns t0 to 0 ns.

3. The image updates live as you move the gate.

7. Display controls

Figure 4. Display panel: **1** min photons/px (blank dim pixels), **2** noise floor (counts, read on the summed decay), **3** floor *auto* (robust baseline), **4** floor *reset* (0 — no subtraction), **5** colormap.

- **min photons/px** masks pixels too dim to trust.
- **noise floor** subtracts a flat background so a pedestal doesn't bias the lifetime. **auto** picks a robust baseline; **reset** turns subtraction off.
- **lock scale** freezes the colour range so planes/frames are comparable.

8. Spatial binning

Figure 5. Binning panel: **1** bin factor, **2** target photons/px, **3** *Auto* (suggest a factor from photon statistics), **4** *reset* (1×, off).

FLIM pixels are often photon-starved. Binning pools each pixel's neighbourhood (`B×B`) for cleaner decays and lifetimes. Click **Auto** to reach the target photon budget, or set the factor by hand. **reset** (↻) returns to 1× (off).

> Binning is the fastest way to make a per-pixel lifetime *map* usable on sparse data — see §9 and §10.

9. Lifetime & phasor analysis

9.1 Two-gate RLD

Press **T** for Lifetime mode. RLD reads an apparent lifetime from two equal-width gates, per pixel, with no fitting:

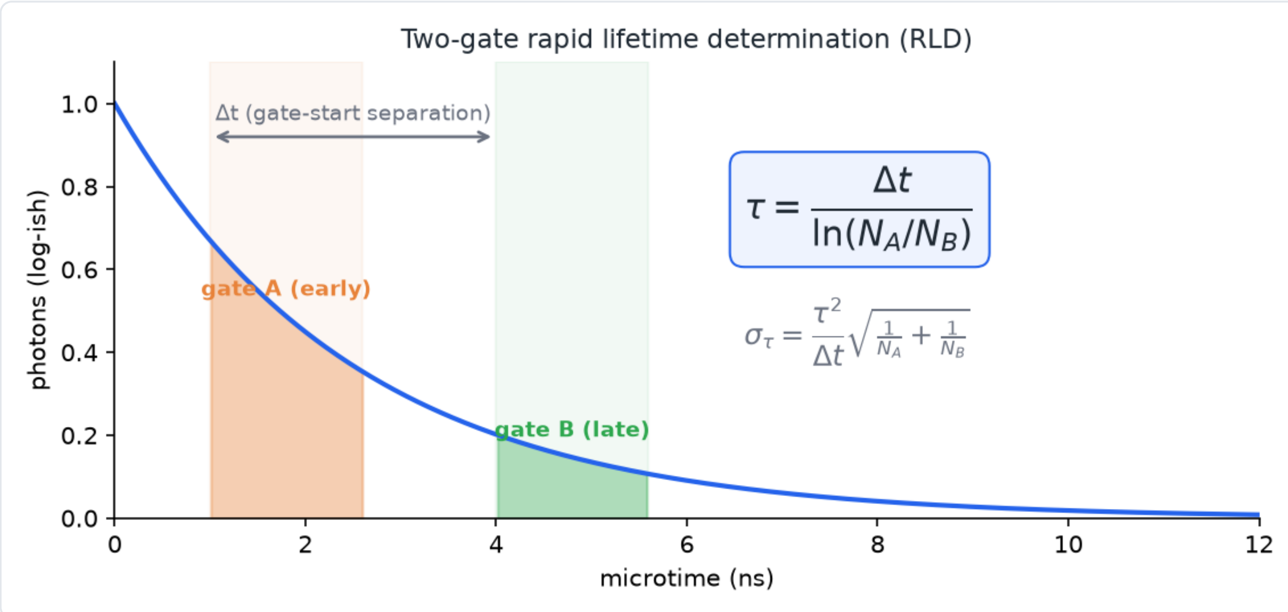

Figure 6. Two-gate RLD. With equal-width gates separated by Δt, the width/amplitude factors cancel, giving τ = Δt / ln(N_A/N_B). The shot-noise uncertainty σ_τ is reported for a pooled region.

1. In the Lifetime panel choose which gate you are editing (**A** early / **B** late) and position the two gates on the decay.
2. Keep the gates **equal width** (the panel warns if not) — RLD assumes it.
3. The τ image appears on the right. Select a region (§11) and open the one-page report (§12) to get the pooled **τ ± σ** (shot-noise uncertainty).

9.2 Phasor

Press **P**. Each pixel's decay becomes a point `(g, s)`; a single lifetime lands on the universal semicircle, mixtures fall inside it. Use **Calibrate phasor from reference…** (View menu) with a dye of known lifetime to remove the instrument/t0 offset, and the **2nd harmonic** toggle to spread short lifetimes apart.

10. IRF reconvolution fit (rigorous lifetime)

For an IRF-deconvolved lifetime (rather than fit-free contrast), use **View ‣ IRF lifetime fit…**.

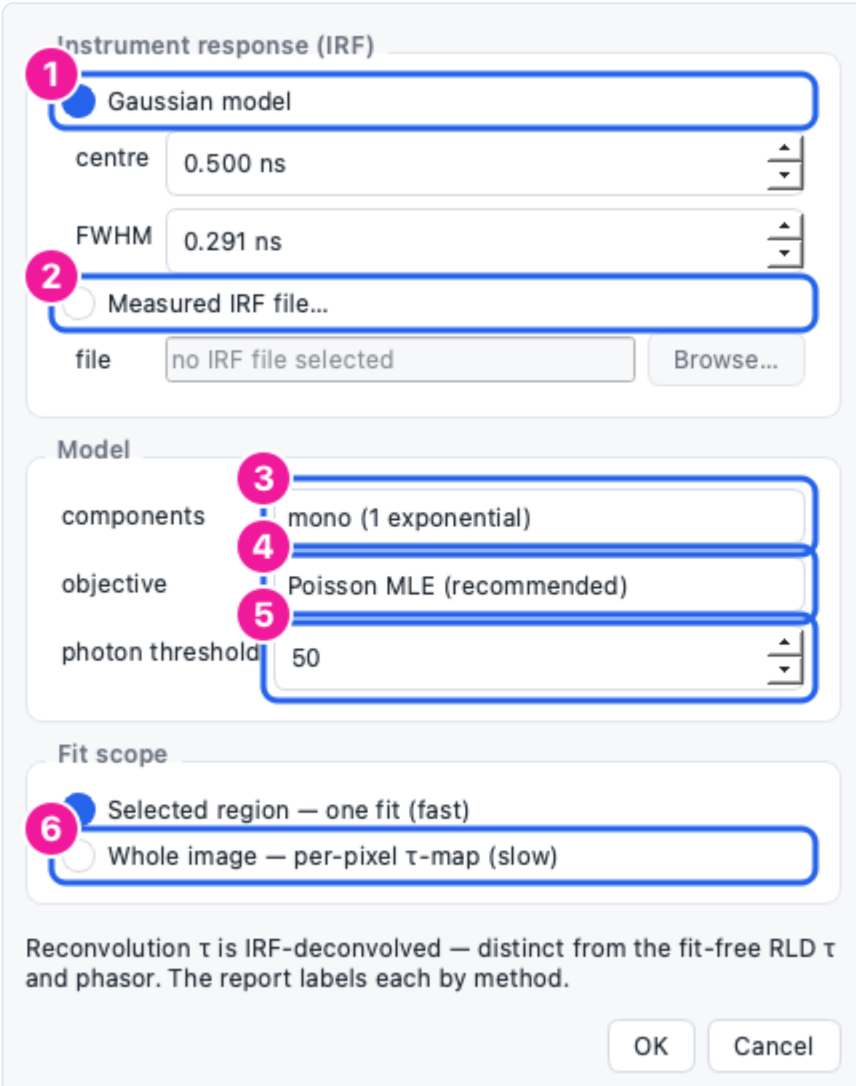

Figure 7. IRF fit dialog: **1** Gaussian IRF model, **2** measured IRF file, **3** components (mono/bi), **4** objective (Poisson MLE / weighted χ^2), **5** photon threshold, **6** whole-image τ-map (vs. a region fit).

1. **Choose the IRF.** For a first look, a **Gaussian** model is fine. For **absolute** lifetimes, load a **measured IRF** file (a scatter/reflection or reference-dye acquisition) — reconvolution is only as good as its IRF.
2. **Model** — mono for one lifetime, bi for two.
3. **Objective** — Poisson MLE (recommended for low counts) or weighted χ^2.
4. **Scope** — a **region fit** (select a bright ROI first) gives one trustworthy $\tau \pm \sigma$; a **whole-image τ-map** fits every pixel above the photon threshold.
5. Read the result: **reduced $\chi^2 \approx 1$** means a good fit. A large χ^2 (e.g. tens or hundreds) means the model is wrong — try a measured IRF, or bi-exponential (see Troubleshooting). A component reported as **σ = n/a (unidentifiable)** is an honest flag that the data cannot separate it.

> For a per-pixel map on sparse data, **bin first** (§8) so pixels clear the photon threshold, then run the map. The default threshold adapts to the image.

11. Selecting pixels & regions

- **Drag a box** on the image for a rectangular ROI; **lasso** on the phasor plot to select a population; or use the **Pixel list** (`View ▸ Pixel list`) to pick by metric.
- Hover to preview a pixel's decay; click to lock it; **Pin** to keep several for comparison.
- A selection travels with every export and scopes the report's statistics.

12. Exporting & the one-page report

Figure 8. Export dialog: **1** raster TIFF (raw values, for ImageJ), **2** one-page report (PNG+PDF with the headline number, decay, image and provenance), **3** restrict to the current selection, **4** *Export & open in Fiji*.

- **File ▸ Export** (`Ctrl+E`) — choose artefacts (raw TIFF, colormapped PNG, decay CSV, per-pixel table, selection files) and a folder. A content-hashed **provenance JSON** is always written.
- **File ▸ Export report (one-pager)…** (`Ctrl+R`) — the one-page summary; in Lifetime mode it includes the pooled **$\tau \pm \sigma$** for the selection.
- **Restrict to selection** masks everything outside the selected pixels.
- **Export & open in Fiji** writes a raw TIFF plus an ImageJ macro (range, LUT, ROI) and launches Fiji on it (set the Fiji path in Preferences first).

13. Troubleshooting

Symptom	Cause	Fix
τ-map fits **0 px**	photon threshold above every pixel's count	bin more (§8), lower the threshold, or do a region fit
IRF fit **reduced $\chi^2 \approx$ 100s**	wrong/guessed IRF, or model too simple	load a measured IRF; try bi-exponential; check the Gaussian centre/FWHM
σ shows **n/a (unidentifiable)**	near-degenerate fit (e.g. two equal τ)	expected/honest — simplify the model or accept the point estimate
`.sdt` won't open (`sdtfile` missing)	environment predates the dependency	relaunch (auto-rebuild) or `pip install -e .`
RLD τ looks wrong	unequal gates, or no background subtraction	make gates equal width; set the noise floor (§7)
app "unidentified developer" warning	installers are unsigned	right-click ‣ Open (macOS) / More info ‣ Run (Windows)
app closes unexpectedly	an unhandled error	a diagnostic log is saved automatically — send it on (see below)

If ChronoGate closes unexpectedly, it writes a dated log naming the fault. Send the newest file from:

- **Windows:** `%LOCALAPPDATA%\ChronoGate\logs`
- **macOS:** `~/Library/Logs/ChronoGate`

(The error dialog shows the exact path. Only the last 20 logs are kept.)

14. Revision history

This SOP is generated from the ChronoGate source and refreshed with each release, so its screenshots and instructions track the current version.

Version	Date	Notes
0.18.1	2026-08-05	Generated from ChronoGate 0.18.1.